



On a generalized two-temperature heat transfer model for non-equilibrium heating in metals with moving heat source



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Highlights

- A generalized two-temperature heat transfer model with a moving heat source was established.
- Convection effects by relative motion were investigated via rigorous mathematical proof.

Typesetting math: 65% characteristic angle within the heating zone was found.

- Peridynamic two-temperature heat model with moving heat sources was developed.

Abstract

Classical one-temperature heat transfer models, whether Fourier or non-Fourier, often overestimate temperature rise and fail to capture the intricate energy exchange and non-equilibrium heating between electrons and phonons in metallic systems. To address this limitation, we introduce a generalized two-temperature heat transfer model that incorporates a moving heat source, emphasizing the convection effects induced by the relative motion between the heat source and the metallic body. Through rigorous mathematical proofs and numerical simulations, we show the hyperbolic characteristics arising from the interplay between the speed of heat transfer and the velocity of the moving heat source. Our findings reveal a strong correlation between the thermal diffusive domain and the heat source's speed, leading to the formation of a distinct characteristic angle within the heating zone. Numerical studies, combining peridynamic spatial discretization with the Crank–Nicolson time marching, are conducted for one-dimensional bars and two-dimensional plates. These simulations highlight the influence of the heat source's velocity and non-Fourier behavior on the overall thermal response.



Keywords

Generalized two-temperature heat transfer; Moving heat source; Non-equilibrium heat process; Peridynamic heat transfer; Electron–phonon interaction; Heat transfer in metals

In modern manufacturing and material processing, precise control of heat conduction processes is a core challenge for ensuring process quality and material performance. Advanced technologies such as laser welding, ultrashort-pulse laser cutting, and metal additive manufacturing [1], [2], [3], [4], characterized by ultrahigh energy densities, ultrafast dynamic heat sources, and multi-timescale non-equilibrium thermal processes, fundamentally challenge the predictive capabilities of classical Fourier-based heat conduction models.

In classical heat conduction theory, Fourier's law linearly correlates heat flux with temperature gradient, yet its instantaneous equilibrium assumption leads to an infinite propagation speed of thermal signals, forming a thermodynamic paradox. This inherent limitation causes significant inaccuracies in extreme transient scenarios like ultrafast laser processing [2]. To address the issue of infinite propagation speed in Fourier-type heat conduction theory, a modified Fourier law was developed as a non-steady-state model by introducing a thermal relaxation parameter τ_q . This leads to hyperbolic partial differential equations that modify heat propagation speed. While this model has been incorrectly attributed to Cattaneo [5] due to a flawed mathematical derivation [6], the correct theoretical foundation was established much earlier by Ref. [7]. Therefore, we refer to this as the modified Fourier law rather than using incorrect historical attribution. Similarly, the Jeffrey-type heat conduction model [8] introduce thermal relaxation τ_q and thermal retardation τ_T into the heat flux and the temperature, respectively. Though these studies have been widely explored in various engineering problems, Bright et al. [9] observed anomalous temperature oscillations in micro-scale experiments, indicating the necessity for deeper theoretical reconstruction of these models. To enhance the understanding of heat transport, the Cattaneo–Fourier (C–F) generalized heat transport model extends non-Fourier theory by employing a multiscale analysis of the Boltzmann Transport Equation. It establishes constitutive relations that encompass ballistic-diffusive spatial scales and finite-infinite speed temporal regimes. The C–F framework incorporates energy stratification: slow C-processes dominate macroscopic diffusion in low-energy regions, while fast F-processes govern microscopic energy transport in high-energy zones [10], [11], [12]. Experimental studies on metallic materials demonstrate that the C–F model improves accuracy in predicting asymmetric temperature distributions during femtosecond laser processing compared to the C–V model [13], [14].

The development of non-equilibrium heat conduction theory based on the two-temperature (2T) heat transfer model has provided critical insights into energy transfer mechanisms between electron–phonon systems [15]. By distinguishing electron temperature (characterizing *electron* energy distribution) from lattice temperature (reflecting phonon thermal states), this model describes their non-equilibrium coupling processes. Remarkable studies regarding the development of the two-temperature heat transfer model, as well as its applications, have been widely investigated. Chen et al. [16] pioneered a

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semiclassical 2T model by replacing the electron heat diffusion equation with the Boltzmann transport equation, significantly improving simulation accuracy for electron drifting effects in ultrafast laser heating. Poletkin et al. [17] further extended phonon heat transfer characterization through the thermal wave model, establishing a hyperbolic heat transfer 2T model that outperforms traditional approaches in simulating electron–phonon coupling in femtosecond-laser-heated gold films.

Building upon the theoretical framework of non-equilibrium heat conduction based on the 2T model, the modeling of dynamic moving heat sources has emerged as a critical challenge in simulating transient thermal processing technologies such as laser welding and additive manufacturing. When heat sources scan material surfaces at high speeds, they induce intense temperature gradients and multiphase transformation coupling effects, which cannot be accurately captured by traditional static heat source models due to spatiotemporal coupling complexities. Weichert et al. [18] systematically investigated three-dimensional temperature field distributions generated by moving heat sources, revealing coupling mechanisms among material thermophysical properties, geometric parameters, and motion velocity. Cheng and Lin [19] developed an analytical model for three-dimensional temperature fields in finite-thickness plates under Gaussian-distributed moving heat sources, establishing mathematical representations of temperature evolution under constant velocity conditions. Ali and Zhang [20] proposed a relativistic heat conduction model (RHCE), achieving the first relativistic covariant description of moving heat source problems while ensuring universal adherence to the second law of thermodynamics. Peterson et al. [21] derived time-varying analytical solutions for two-dimensional moving laser heat sources, revealing dimensional differences in heat conduction through planar effect analysis.

In the realm of numerical modeling for the thermal response subjected to moving sources, various studies have been conducted. A new approach for numerical calculation in the field of diffusion–convection problems was provided by Adam Bokota and Sławomir Iskierka [22] through their study of the finite element method for solving such problems in the presence of a moving heat point source. Shuja and Yilbas [23] employed finite element methods to uncover dynamic responses of coupled temperature–stress fields in steel plates under multi-beam laser irradiation. J. Kidawa-Kukla [24] established exact analytical solutions for temperature fields in rectangular plates heated by moving sources using the Green’s function method, validated through numerical experiments. Ma et al. [25] focused on unsteady thermal stress effects induced by double-elliptical laser distributions on moving steel plates, elucidating heat transfer regulation mechanisms through spot positioning and intensity modulation. Satish K. Velaga and A. Ravisankar [26] conducted a finite element-based parametric study on the geometric parameters in regulating weld pool morphology, thermal gradients, and residual stress distributions, thereby providing

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theoretical support for precise calibration of heat source geometry in welding simulations Juan et al. [27] developed a hybrid element-free Galerkin-finite element algorithm, significantly enhancing computational efficiency and accuracy for transient heat transfer problems with concentrated moving sources. These numerical methods have demonstrated their effectiveness in predicting the heat transfer processes and thermal responses of continuum bodies. However, they face significant challenges when applied to cracked materials or when size effects come into play, given that these methods are formulated on the classical local heat transfer theory that is expressed by partial differential equations.

In contrast, a nonlocal methodology, namely peridynamics, has been designed to address the nonlocal effects and discontinuities inherent in material behavior. It encompasses bond-based and state-based peridynamics [28], [29], [30], [31], [32]. While bond-based peridynamics [33] was the first, its limitations, like independent bond responses, constrain Poisson's ratios [28], [33]. State-based peridynamics [34], including ordinary and non-ordinary formulations, evolved to overcome these constraints. Numerical methodologies developed for mechanics have also been adapted to peridynamic heat transfer. Bond-based peridynamic heat transfer formulations [29], [35] were introduced, assuming micro-scale heat conductivity. State-based peridynamic heat conduction formulations were proposed in [30], and generalized peridynamic formulations for coupled fields, such as thermoelasticity [36] and poroelasticity [37], were further developed. While research has integrated peridynamics with non-Fourier heat transfer constitutive models [38], [39], [40] and explored non-Fourier heat transfer processes in different materials, peridynamic heat transfer formulations have yet to achieve widespread adoption compared to mechanics applications.

In this work, inspired by the hyperbolic transport equations for local nonequilibrium media and high-velocity traveling thermal waves [41], and building upon the concept of thermal shock waves induced by a moving heat source [42], we propose a generalized two-temperature heat transfer model that incorporates the effects of moving source convection with the application of C–F heat flux model. Similar to the approach in [41] which studied temperature fields around moving heat sources using hyperbolic heat equations, our model analyzes various heat transfer scenarios, accounting for different relative velocities between the metallic body and the moving heat source.

2. Theory

This section systematically elaborates the generalized two- temperature theory constructed upon the C–F heat transfer framework [43] [44]. Following this theoretical development, we incorporate a moving heat source into the model framework

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and systematically investigate the effects of its motion velocity on thermal conduction characteristics described by the proposed generalized two-temperature model.

The generalized two-temperature heat transfer model and its extension to moving heat source problems are formulated based on the C–F heat flux model [43], [44]. In metallic materials subjected to ultra-fast laser heating, the thermal energy transfer mechanism exhibits a distinct two-stage process: initial absorption by the electron subsystem, followed by energy transmission to the phonon subsystem through electron–phonon interactions. Based on the seminal two-temperature formulation established by Anisimov et al. [15] for metallic heat transfer processes, the governing equations can be expressed as:

$$\begin{aligned} C_e \frac{\partial T_e}{\partial t} + \nabla \cdot \mathbf{q}_e &= -G (T_e - T_l) + S \\ C_l \frac{\partial T_l}{\partial t} &= G (T_e - T_l) \end{aligned} \quad (1)$$

where the subscripts ‘*e*’ and ‘*l*’ represent the properties in the electron process and the phonon process, respectively; *G* is a coupling coefficient that links these two processes with the unit of J/K s; *C_e* and *C_l* represent the heat capacities with respect to electron and lattice, respectively; *S* is the external heating due to laser pulse. By introducing different types of heat flux models into $\mathbf{q}_e$, the governing Eq. (1) yields the corresponding two-temperature theories, such as the parabolic two-temperature model and the hyperbolic two-temperature model [41]. The hyperbolic approach was first introduced in [41] and further developed through a discrete heat equation formulation in [45]. Furthermore, a nonlocal two-temperature model for metals irradiated by ultrashort laser pulses was developed in [46].

2.1. Generalized C–F heat flux model

In the present work, the generalized C–F heat transfer model was proposed to describe the heat transfer process of electrons. The generalized heat flux model, C–F heat flux model was originated from the fundamental Boltzmann Transport Equation (BTE) under the assumption that both the slow heat transfer process and the fast heat transfer process exist in the system simultaneously. To support this assumption, the distribution of the heat carrier at different frequencies is given in Fig. 1, which indicates that the high frequency motion (fast process) and low frequency motion (slow process). In coincidence with their associated physical characteristics, the thermal signal propagates at an infinite speed in the **F**-type heat transfer process, while the heat propagates in a wave-like way with a finite speed in the **C**-type heat transfer process.

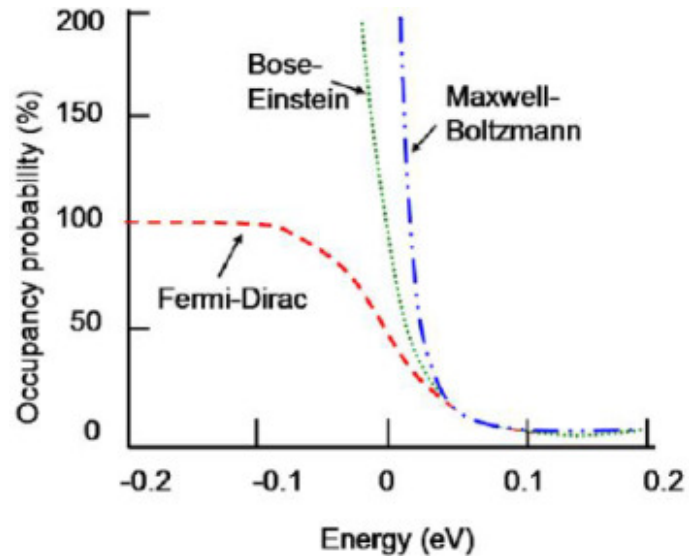
The flux due to heat conduction of electron heat carriers is then given as follows:

$$\begin{aligned} \mathbf{q}_e(\mathbf{x}) &:= \int_0^{\omega_D} \mathbf{v}(\mathbf{x}) f(\mathbf{x}) \bar{h}\omega D(\omega) d\omega \\ &= \int_0^{\omega_T} \mathbf{v}(\mathbf{x}) f(\mathbf{x}) \bar{h}\omega D(\omega) d\omega + \int_{\omega_T}^{\omega_D} \mathbf{v}(\mathbf{x}) f(\mathbf{x}) \bar{h}\omega D(\omega) d\omega \\ &= \mathbf{q}_{C_e} + \mathbf{q}_{F_e} \end{aligned} \quad (2)$$

where $\mathbf{f}(\mathbf{x})$ is the heat carrier's distribution and is a function of energy; ω represents heat carrier's vibration frequency and ω_T is a threshold frequency that separates the low frequency and high frequency motion; $\bar{h}$ is Planck's number divided by 2π and $\bar{h}\omega$ is the energy; $D(\omega)$ is the density of states. The assumption herein is that the integral up to a threshold frequency ω_T associates with the slow process ($\mathbf{q}_{C_e}$) and that the integral from the threshold to the Debye frequency, ω_D associates the fast process ($\mathbf{q}_{F_e}$). In order to obtain the $\mathbf{q}_{C_e}$ and $\mathbf{q}_{F_e}$, multiplying the transient BTE by $\mathbf{v}\psi$, where $\psi = \bar{h}\omega D(\omega)$, and integrating over the entire frequency ranges (0 to ω_D) yields the following:

$$\begin{aligned} &\int_0^{\omega_T} \mathbf{v}\psi \frac{\partial f}{\partial t} d\omega + \int_0^{\omega_T} \mathbf{v} \cdot \mathbf{v} \nabla f \psi d\omega + \int_{\omega_T}^{\omega_D} \mathbf{v}\psi \frac{\partial f}{\partial t} d\omega + \int_{\omega_T}^{\omega_D} \mathbf{v} \cdot \mathbf{v} \nabla f \psi d\omega \\ &= \underbrace{\int_0^{\omega_T} \mathbf{v}\psi \frac{f_0 - f}{\tau} d\omega + \int_{\omega_T}^{\omega_D} \mathbf{v}\psi \frac{f_0 - f}{\tau} d\omega}_{\int_0^{\omega_D} \left(\frac{\partial f}{\partial t} \right)_{scatt} d\omega} \end{aligned} \quad (3)$$

where $\mathbf{a}\partial f/\partial \mathbf{v}$ in the original BTE is omitted and the linear approximation is introduced to deal with the scattering term. In the high-frequency domain, as can be observed in Fig. 1, the distribution of function $\mathbf{f}$ can be considered as a constant; hence $\partial f/\partial t = 0$. Consequently, Eq. (3) can be degenerated into two equations based on the associated frequency regimes:



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Fig. 1. Occupancy probability for the different distributions.

Figure adapted from [43], [47].

1. **Slow motion:** low frequency ($0 \rightarrow \omega_T$)

$$\int_0^{\omega_T} \mathbf{v} \psi \frac{\partial f}{\partial t} d\omega + \int_0^{\omega_T} \mathbf{v} \cdot \mathbf{v} \nabla f \psi d\omega = \int_0^{\omega_T} \mathbf{v} \psi \frac{f_0 - f}{\tau} d\omega \quad (4)$$

After mathematical manipulations, we have the following:

$$\mathbf{q}_{C_e} + \tau \frac{\partial \mathbf{q}_{C_e}}{\partial t} = -k_C \nabla T = -(1 - F_T) K \nabla T \quad (5)$$

2. **Fast motion:** high frequency ($\omega_T \rightarrow \omega_D$)

$$\int_{\omega_T}^{\omega_D} \mathbf{v} \cdot \mathbf{v} \nabla f \psi d\omega = \int_{\omega_T}^{\omega_D} \mathbf{v} \psi \frac{f_0 - f}{\tau} d\omega \quad (6)$$

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After mathematical manipulations, we have

$$\mathbf{q}_{F_e} = -\mathbf{k}_F \nabla T = -F_T K \nabla T \quad (7)$$

where K and F_T are defined as follows:

$$K = \int_0^{\omega_T} \mathbf{v} \cdot \mathbf{v} \frac{df_0}{dT} \psi d\omega + \int_{\omega_T}^{\omega_D} \mathbf{v} \cdot \mathbf{v} \frac{df_0}{dT} \psi d\omega = K_C + K_F \quad (8)$$

$$F_T = \frac{\int_{\omega_T}^{\omega_D} \mathbf{v} \cdot \mathbf{v} \frac{df_0}{dT} \psi d\omega}{\int_0^{\omega_D} \mathbf{v} \cdot \mathbf{v} \frac{df_0}{dT} \psi d\omega} = \frac{K_F}{K_F + K_C}$$

Therefore, the generalized C–F thermal flux is given by

$$\begin{aligned} \mathbf{q}_e &= \mathbf{q}_{F_e} + \mathbf{q}_{C_e} \\ \mathbf{q}_{F_e} &= -\mathbf{k}_e F_T \nabla T_e \\ \mathbf{q}_{C_e} + \tau \frac{\partial \mathbf{q}_{C_e}}{\partial t} &= -(1 - F_T) \mathbf{k}_e \nabla T_e \end{aligned} \quad (9)$$

where F_T represents the heat conduction model number, $F_T \in [0, 1]$.

Remark 2.1

It should be noted that the slow and fast processes may have different physical interpretations in other studies; however, further discussion of these alternative interpretations is beyond the scope of the present work. Derived from the fundamental C–F model, the two-temperature heat transfer models can be systematically characterized through the parametric pair (F_T, τ) , including the Fourier, modified Fourier, and Jeffreys'-type heat conduction equations. In specifics:

1. $F_T = 1, \tau = 0$ The generalized C–F flux model degenerates to the Fourier heat flux model as follows

$$\mathbf{q}_e = -\mathbf{k}_e \nabla T_e \quad (10)$$

The resulting model recovers the parabolic two-temperature model, where the speed of the heat transfer process is infinite.

2. $F_T = 0, \tau \neq 0$ The generalized C–F flux model degenerates to the modified Fourier heat flux model [7] as follows

$$\mathbf{q}_e + \tau \dot{\mathbf{q}}_e = -k_e \nabla T_e \quad (11)$$

The resulting model recovers the hyperbolic two-temperature model.

3. $F_T \in (0, 1), \tau \neq 0$: Jeffreys'-type heat flux In this case, the generalized C–F flux model degenerates to the Jeffreys'-type heat flux model.

$$\mathbf{q}_e + \tau \dot{\mathbf{q}}_e = -k_e (\nabla T_e + \tau F_T \nabla \dot{T}_e) \quad (12)$$

2.2. Generalized two-temperature heat transfer model with a moving heat source

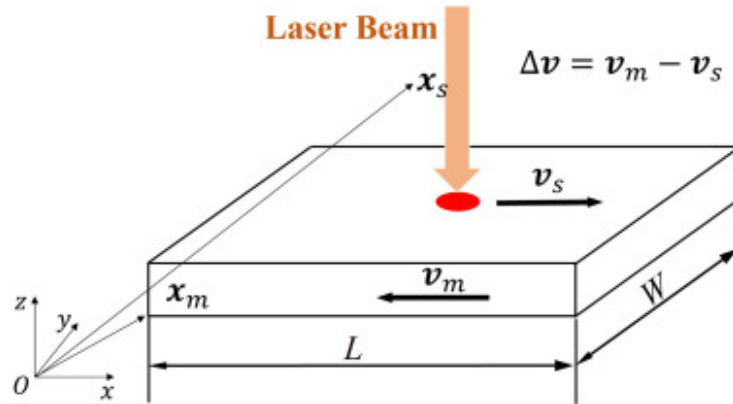
Existing studies regarding general heat transfer modeling with a moving heat source usually consider the heat source as a moving flux boundary; the in-depth relationship between the moving speed and diffusion process is mostly studied via observations [48]. In contrast, the present work considers a stationary coordinate system with a relative velocity $\Delta \mathbf{v}$ between the metallic body and the heat source. When there is no relative motion ($\Delta \mathbf{v} = 0$), the two-temperature heat transfer model is given in Eq. (1). However, cases with non-zero relative velocity ($\Delta \mathbf{v} \neq 0$) must consider the 'convection' effect due to the relative motion between the heat source and the material. The resulting generalized heat conduction model for the moving heat source is given as follows:

$$\begin{aligned} C_e \left(\frac{\partial T_e}{\partial t} + \nabla \cdot (\Delta \mathbf{v} T_e) \right) + \nabla \cdot \mathbf{q}_e &= -G (T_e - T_l) + S \\ C_l \frac{\partial T_l}{\partial t} + C_l \nabla \cdot (\Delta \mathbf{v} T_l) &= G (T_e - T_l) \\ \mathbf{q}_e &= \mathbf{q}_{F_e} + \mathbf{q}_{C_e} \\ \mathbf{q}_{F_e} &= -F_T k_e \nabla T_e \\ \mathbf{q}_{C_e} + \tau \left[\frac{\partial \mathbf{q}_{C_e}}{\partial t} + \nabla \cdot (\Delta \mathbf{v} \otimes \mathbf{q}_{C_e}) \right] &= -(1 - F_T) k_e \nabla T_e \end{aligned} \quad (13)$$

where $\Delta \mathbf{v}$ represents the relative velocity between metal body of interest ($\mathbf{v}_m$) and the moving heat source ($\mathbf{v}_s$), and

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In the present work, we fix the metal object/workpiece of interest in the configuration ($\mathbf{v}_m = \mathbf{0}$), leading $\Delta\mathbf{v} = -\mathbf{v}_s$.



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Fig. 2. A schematic view of the laser moving heat source, workpiece movement, coordinate systems, and relative velocity. $\mathbf{x}_s$ and $\mathbf{x}_m$ represent the positions of the laser beam and workpiece in the stationary coordinate system $O - xyz$, respectively. $\mathbf{v}_s$ and $\mathbf{v}_m$ denote the absolute velocities of the laser beam and workpiece, respectively. The relative velocity between the laser beam and workpiece is given by $\Delta\mathbf{v} = \mathbf{v}_m - \mathbf{v}_s$. Note that the directions of $\mathbf{v}_m$ and $\mathbf{v}_s$ can be flexible.

Remark 2.2

1. Eq. (13) explicitly incorporates the velocities $\mathbf{v}_m$ and $\mathbf{v}_s$, diverging from previous studies that treated the heat source as a boundary condition. This formulation offers a compact mathematical model for analyzing the interplay between the diffusion process and the moving heat source. Further details are provided in Section 2.3.
2. The proposed model agrees with the replacement as mentioned in Ref. [49], whereas it indicates that the influence of moving heat source is only on the slow heat process, $\mathbf{q}_C$.
3. The proposed model naturally inherits the physical features of the C-F heat conduction model of bridging the transition with different scales in both space (from the ballistic to diffusive heat conduction) and time (from the finite wave speed to infinite wave speed). More specifics can be found in Refs. [11], [43], [50].

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4. The proposed model can be converted to a general convection–diffusion formulation as the moving velocity-related term is a convection term in physics. Tuning $F_T = 0$ yields a hyperbolic convection–diffusion model in Ref. [51].

2.3. Hyperbolic characteristics of the proposed two-temperature heat conduction model

In this section, we follow the derivation proposed in Refs. [11], [52], [53] regarding the hyperbolic characteristics of the one-temperature heat conduction model and extend it to the proposed two-temperature model, the Galilean invariance of the proposed two-temperature heat model the proposed model is further investigated. We consider the one-dimensional hybrid form of Eq. (13), and rewrite it as follows:

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{H} = \mathbf{W} \quad (14)$$

where $\mathbf{U}$ denotes the state vector, matrix $\mathbf{H}$ denotes the flux, and $\mathbf{W}$ is the vector of source terms. The specifics are given as follows:

$$\mathbf{U} = \begin{bmatrix} C_e T_e \\ C_l T_l \\ 0 \\ \tau q_{Ce} \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} C_e \Delta v T_e + q_{Ce} + q_{Fe} \\ C_l \Delta v T_l \\ F_T k_e T_e \\ (1 - F_T) k_e T_e + \Delta v \tau q_{Ce} \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} -G(T_e - T_l) + S \\ G(T_e - T_l) \\ -q_{Fe} \\ -q_{Ce} \end{bmatrix} \quad (15)$$

Using the chain rule gives:

$$\nabla \cdot \mathbf{H} = \frac{\partial \mathbf{H}}{\partial \mathbf{U}} \frac{\partial \mathbf{U}}{\partial X} = \mathbf{A} \frac{\partial \mathbf{U}}{\partial X} \quad (16)$$

where $\mathbf{A}$ is the Jacobian of $\mathbf{H}$ with respect to $\mathbf{U}$, and is given as follows:

$$\mathbf{A} = \begin{bmatrix} \Delta v, & 0, & 0, & 1/\tau \\ 0, & \Delta v, & 0, & 0 \\ F_T k_e / C_e, & 0, & 0, & 0 \\ q_{Fe} / C_e, & 0, & 0, & \Delta v \end{bmatrix} \quad (17)$$

Solving the eigenvalue problem with respect to an arbitrary unit vector $\boldsymbol{\omega}$ with $\|\boldsymbol{\omega}\| = 1$:

$$\det(\lambda_k \mathbf{I} - \mathbf{A} \cdot \boldsymbol{\omega}) = 0 \quad (18)$$

yields the following four real eigenvalues λ_k for the 1D case:

$$\begin{aligned} \lambda_1 &= 0, \quad \lambda_2 = \Delta v, \\ \lambda_3 &= \Delta v + \sqrt{\frac{k_e(1-F_T)}{C_e \tau}}, \quad \lambda_4 = \Delta v - \sqrt{\frac{k_e(1-F_T)}{C_e \tau}} \end{aligned} \quad (19)$$

The above indicates that there exist two characteristic speeds in the proposed two-temperature model, one is the relative velocity (Δv), i.e the speed of moving heat source in the present work, and the other is the following propagation speed of diffusion

$$c = \sqrt{\frac{k_e(1-F_T)}{C_e \tau}} \quad (20)$$

which is also referred to as the second sound speed in the area of heat transfer [54]. Notably, when $F_T = 0$, the diffusion speed c reverts to the classical wave speed of hyperbolic diffusion.

It can be further concluded that the thermal response during moving heat source processes (e.g., laser heating) is fundamentally governed by the interplay between the above two characteristic velocities: the relative speed Δv and the thermal diffusion speed c . To better characterize this, we define the following non-dimensional number:

$$H_c = \frac{\|\Delta v\|}{c} \quad (21)$$

which can be used to classify the various diffusion types. Specifically:

$$\begin{aligned} H_c &< 1 \Leftrightarrow \text{Sub-critical heat transfer,} \\ H_c &> 1 \Leftrightarrow \text{Super-critical heat transfer,} \\ H_c &= 1 \Leftrightarrow \text{Critical heat transfer.} \end{aligned} \quad (22)$$

Remark 2.3

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The hyperbolic characteristic number H_c plays a role in heat transfer analogous to the Mach number in compressible fluid dynamics, characterizing the propagation of disturbances within a medium [52]. It is worthy noting the temperature field around moving heat source described by the pure hyperbolic heat equation was first considered in pioneering study [41]. In this work, the laser heat source is modeled as a thermal disturbance propagating through the solid. The analogy reveals distinct heat diffusion regimes along with the moving heat source:

1. **Pure Fourier case** ($F_T = 1, \tau = 0$): Sub-critical heat transfer permits bidirectional (upstream and downstream) propagation, while super-critical diffusion confines propagation to downstream only when the material velocity $\|\Delta \mathbf{v}\|$ exceeds the diffusive speed c .
2. **Non-Fourier case** ($F_T \neq 1, \tau \neq 0$): In this hybrid regime, the propagation mechanism combines Fourier and Cattaneo-type behaviors, manifesting as a distinctive conical pattern of elevated concentration with characteristic bounding angles, while maintaining bidirectional propagation capability. The specific formulation of the angle of “conical regime” is characterized as follows:

$$\alpha = \sin^{-1} \left(\frac{1}{H_c} \right) \quad (23)$$

A related concept, termed the “thermal shock angle” was introduced in [42], which is a particular case of ($F_T = 0, \tau \neq 0$) in the present work.

3. **Galilean invariance** [53]: The independence of the diffusive speed c from both heat source kinematics and system configuration demonstrates that the proposed two-temperature model inherently preserves Galilean invariance under moving heat source conditions.

3. Numerical discretization

We now present the specific numerical discretization in both spatial and temporal dimensions for the proposed two-temperature model subjected to a moving heat source. We utilize the newly developed Peridynamic differential operator [55] to spatially discretize the differential operators within Eq. (13). After the spatial semi-discretization, the time-continuous ordinary

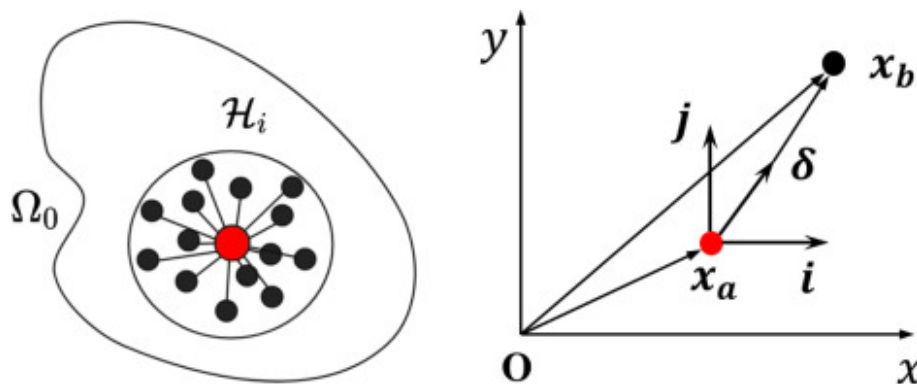
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differential equations (see Eq. (30)) are obtained for time marching, which is discretized by the Second-order time-accurate Crank–Nicolson scheme.

3.1. Peridynamic differential operators

Peridynamics, as a nonlocal reformulation of solid mechanics, has gained much popularity in computational mechanics in the last two decades. The PD equation of motion is expressed in an integral–differential form, characterized by an inherent length scale known as the “horizon”. Within this framework, material points interact with one another within the domain defined by the horizon.

The inherent simplicity and nonlocal nature of PD render it particularly effective for addressing problems involving moving discontinuities, such as fractures, without necessitating specialized treatment. Numerous notable studies have been conducted across a variety of problem domains; however, a detailed discussion of these studies is beyond the scope of the present work. For further insights, readers are referred to [56]. For the sake of simplicity, we herein directly present the following non-local discretized expressions associated with gradient and divergence with respect to general physical quantities ϕ (scalars, such as temperature) and Φ (vectors, such as heat flux), which play a crucial role in the discretization of Eq. (13).



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Fig. 3. The peridynamic description of a thermal body (left) and a thermal bond between x_a and x_b (right) in $2D$.

Consider a domain in Euclidean space, $\mathbb{E}$, as shown in Fig. 3. Let $\mathbf{x}_i \in \Omega_0 \subset \mathbb{E}$ denotes the *geometric* position vector of a certain position in the domain and $\mathbf{x}_j \in \Omega_0 \subset \mathbb{E}$ denotes another point which is close to the position $\mathbf{x}_i$. The peridynamic differential operators $\nabla\phi$ and $\nabla\cdot\Phi$ are given as follows:.

$$\nabla\phi_i := \int_{H_{x_i}} \phi_{ij} x_{ij} \omega(\|x_{ij}\|) dV_j K_i \nabla\cdot\Phi_i := - \int_{H_{x_i}} \Phi_i^T K_i + \Phi_j^T K_j x_{ij} \omega(\|x_{ij}\|) dV_j \quad (24)$$

where i denotes the particle label and H_i is particle i 's horizon; j denotes the particles located within the horizon H_i , $j \in H_i$. $\phi_{ij} := \phi_j - \phi_i$ and x_{ij} is the distance vector pointing from j to i , $x_j - x_i$. The weight function $\omega(\|x_{ij}\|)$ in the present work is taken as follows:

$$\omega(\|x_{ij}\|) = \begin{cases} 1, & \|x_{ij}\| \leq \delta \\ 0, & \|x_{ij}\| > \delta \text{ or material damages} \end{cases} \quad (25)$$

where δ is the size of the horizon. The K_a ($a = i$ or j) tensor is associated with the inverse of the shape tensor of particle a in the peridynamic differential operators [12], [55], [57], as follows:

$$K_a := \int_{H_{x_a}} x_{ab} \otimes x_{ab} \omega(\|x_{ab}\|) dV_j^{-1} \quad (26)$$

where particle b locates within the horizon of particle a . In the present work, we discretize the continuum body by particles such that the integral expressions of $\nabla\phi_i$, $\nabla\cdot\Phi_i$, and K_a in Eq. (24) can be directly evaluated by the following summation forms:

$$\begin{aligned} \nabla\phi_i &:= \sum_{j \in H_{x_i}} \phi_{ij} x_{ij} \omega_{ij} V_j K_i = G^x, G^y [\phi] \nabla\cdot\Phi_i := - \sum_{j \in H_{x_i}} \Phi_i^T K_i + \Phi_j^T K_j x_{ij} \omega_{ij} V_j = D^x, D^y [\Phi] \\ K_a &:= \sum_{b \in H_{x_a}} x_{ab} \otimes x_{ab} \omega_{ab} V_j^{-1} \end{aligned} \quad (27)$$

By observation, in a general two-dimensional case with a total number of particles Ng , the sizes of $[\phi]$ and $[\Phi]$ are $Ng \times 1$ and $2Ng \times 1$. We can extract the coefficient matrices as follows:

$$\begin{aligned} G_{ij}^x &:= \begin{cases} \omega_{ij} V_j x_{ij}^x K_i(1,1) + x_{ij}^y K_i(1,2), & i \neq j \\ - \sum_{j \in H_{x_i}} D_{ij}^x, & i = j \end{cases} \\ G_{ij}^y &:= \begin{cases} \omega_{ij} V_j x_{ij}^x K_i(2,1) + x_{ij}^y K_i(2,2), & i \neq j \\ - \sum_{j \in H_{x_i}} D_{ij}^y, & i = j \end{cases} \end{aligned} \quad (28)$$

and

$$\begin{aligned} D_{ij}^x &:= \begin{cases} \omega_{ij} V_j x_{ij}^x K_j(1,1) + x_{ij}^y K_j(1,2), & i \neq j \\ \sum_{j \in H_{x_i}} \omega_{ij} V_j x_{ij}^x K_i(1,1) + x_{ij}^y K_i(1,2), & i = j \end{cases} \\ D_{ij}^y &:= \begin{cases} \omega_{ij} V_j x_{ij}^x K_j(2,1) + x_{ij}^y K_j(2,2), & i \neq j \\ \sum_{j \in H_{x_i}} \omega_{ij} V_j x_{ij}^x K_i(2,1) + x_{ij}^y K_i(2,2), & i = j \end{cases} \end{aligned} \quad (29)$$

We define $G_{2Ng \times Ng} = G^x, G^{y^T}$ and $D_{Ng \times 2Ng} = D^x, D^y$. Via simple mathematical manipulation, one yields the following time-continuous ordinary differential equations:

$$\dot{M}U(t) + KU(t) = F(t) \quad (30)$$

where the primary unknowns $U(t)$ and its associated changing rate $\dot{U}(t)$ are defined as follows:

$$\begin{aligned} U(t) &= T_e q_{C_e} T_l^T \\ \dot{U}(t) &= \dot{T}_e \dot{q}_{C_e} \dot{T}_l^T \end{aligned} \quad (31)$$

where T_e and q_{C_e} represent the electron field vectors in the metal domain; T_l represent the phonon field vectors in the metal domain. The specific matrices M , K and F are given by

$$\begin{aligned} M &= \begin{pmatrix} C_e I & O & O \\ O & \tau I & O \\ O & O & C_l I \end{pmatrix}, \quad F = S^T \\ K &= \begin{pmatrix} \Delta v \cdot G - F_T k_e D \cdot G + GI & D & -GI \\ (1 - F_T) k_e G & I + \tau \Delta v \cdot G & O \\ -GI & O & \Delta v \cdot G + G + GI \end{pmatrix} \end{aligned} \quad (32)$$

where matrices O and I are zero and identity matrices, respectively; S is the external heat source/sink.

In the present work, the ultrafast Gaussian laser beam [58] is considered as the moving heat source imposed on the metal surface that stimulates the electron–phonon interaction system (see Fig. 4):

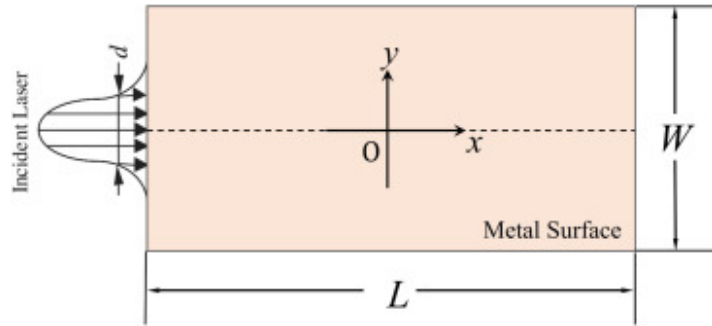
$$q|_m \cdot \underline{n}|_m = Q_0 e^{-8(\Delta x^2/d^2)} \quad (33)$$

Typesetting math: 65% $-x_s \parallel$, x_s denotes the location of the moving heat source center, with a constant moving velocity v_s . d denotes the radius of the laser beam. The $q|_m$ and $n|_m$ represent the heat flux and normal direction of the metal surface that is subjected to

the laser beam. To impose the above heat flux into the proposed model, the right-hand side term S in Eq. (32) is introduced as an external source term [30]:

$$S = \frac{1}{\Delta} Q_0 e^{-8(\Delta x^2 / d^2)} \quad (34)$$

where Δ is taken as the radius of the horizon.



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Fig. 4. Ultrafast laser beam with Gaussian distribution.

3.2. Time marching

Let $U(t) : I := [t_0, t_f] \subset \mathbb{R} \rightarrow \mathbb{R}^{n_{dim}}$ be the primary unknown vector where t_0 and t_f are the initial and final time of simulation, respectively. Consider the temporal discretization of the governing equations for a time interval $I := [t_0, t_f]$ split into sub-intervals, i.e., $I := [t_0, t_f] = \bigcup_{n=0}^{f-1} [t_n, t_{n+1}]$. The time step size is defined as $\Delta t_{n+1} := t_{n+1} - t_n > 0$, and it is assumed to be constant for simplicity, i.e., $\Delta t = \Delta t_{n+1}$. In the present work, we utilized the following implicit Crank–Nicolson scheme to temporally discretize Eq. (30):

$$M \frac{U^{n+1} - U^n}{\Delta t} + \frac{1}{2} K U^{n+1} + U^n = \frac{1}{2} F^{n+1} + F^n \quad (35)$$

and

$$\frac{1}{2} F^{n+1} + F^n + M U^n - \frac{\Delta t}{2} K U^n \quad (36)$$

Solving the Eq. (36) yields U^{n+1} . It is noteworthy that the Crank–Nicolson scheme is a non-dissipative scheme; this is in such a way, the inherent hyperbolic characteristic in the non-Fourier heat transfer may lead to numerical oscillations in the solution. One promising technique to address this issue is to introduce numerical dissipation into the time stepping, however, the dissipative schemes may smooth out the wave and wave-like properties in the heat transfer process. For more information, one is referred to [59], [60], [61].

Additionally, we consider the later beam moving with a constant velocity in the present work, and the position of the moving laser beam x_s is evaluated by:

$$x_s^{n+1} = x_s^n + v_s \Delta t \quad (37)$$

4. Results and discussions

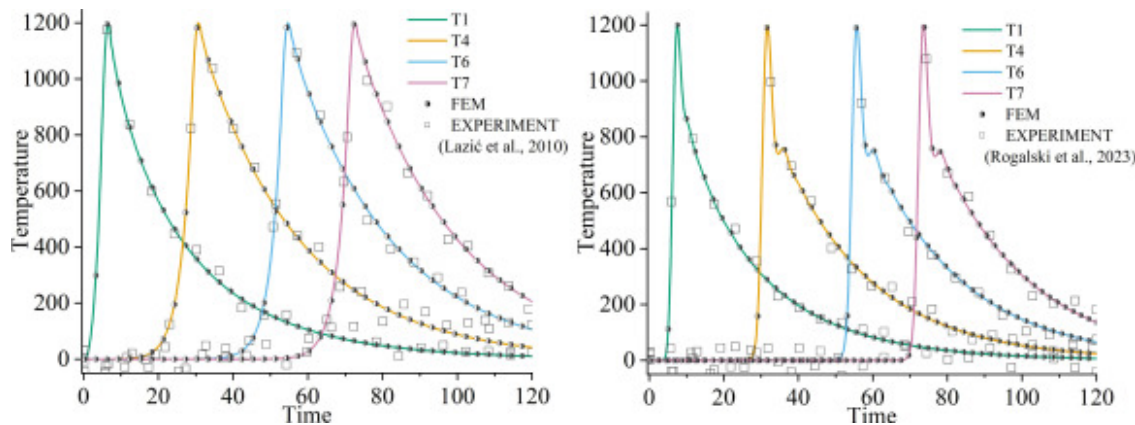
In this section, several numerical examples are exploited with the objective of validating the theory introduced in the preceding sections. In all cases, local boundary conditions, including the moving heat sources, utilize the implementation proposed in Ref. [30] in the proposed computational framework. We compare numerical results with experimental data to verify the accuracy of the proposed method. We then demonstrate the constrained heat transfer area and the specific angle resulting from the competition between wave speed and the speed of the moving heat source via a 2D metallic plate, thereby validating the hyperbolic characteristics of the proposed two-temperature heat conduction model.

4.1. Numerical validation: heat process in a one-dimensional bar with moving heat source

To validate the proposed generalized two-temperature peridynamic model, we perform systematic numerical simulations by replicating welding experiments conducted by Lazić et al. [62], a representative engineering scenario involving a moving heat source. Four points along the trajectory of the heat source are selected, and the corresponding temperature histories at these positions are presented in Fig. 5. When the parameter $H_c = 10^{-5}$ (Fig. 5, left), the resulting high wave velocity approximates Fourier-type heat conduction (Eq. (22)). This is evidenced by the close agreement between the simulated temperature curves and the Fourier-based experimental results reported in Ref. [62]. Conversely, for $H_c = 1$ (Fig. 5, right), the numerical simulation captures the non-equilibrium transient behavior observed experimentally by Rogalski et al. [63], thereby confirming the model's ability to accurately represent rapid thermal interactions.

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The specific material parameters [62] is given as: density-specific heat product $\rho c_p = 4\,898\,556 \text{ J/m}^3 \text{ }^\circ\text{C}$, thermal conductivity $k = 36 \text{ W/m }^\circ\text{C}$, and heat source power $Q = 4256 \text{ W}$. Boundary conditions include initial/ambient temperatures $T_0 = T_\infty = 0 \text{ }^\circ\text{C}$, heat transfer coefficient $h = 10 \text{ W/m}^2 \text{ }^\circ\text{C}$, heat source velocity $v_x = 1.61 \text{ mm/s}$, and time step $\Delta t = 0.1 \text{ s}$ and 500 particles are used for spatial discretization. Additionally, the finite element method with second-order shape functions is also utilized to spatially discretize Eq. (13), aiming to further validate the proposed peridynamic two-temperature model. A good agreement can also be observed in both cases shown in Fig. 5.



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Fig. 5. Time history of temperature at different positions. Left: $H_c = 10^{-5}$ (sub-critical diffusion) and Right: $H_c = 1$ (critical diffusion).

4.2. Hyperbolic characteristics: heat process in a two-dimensional plate with different F_T s

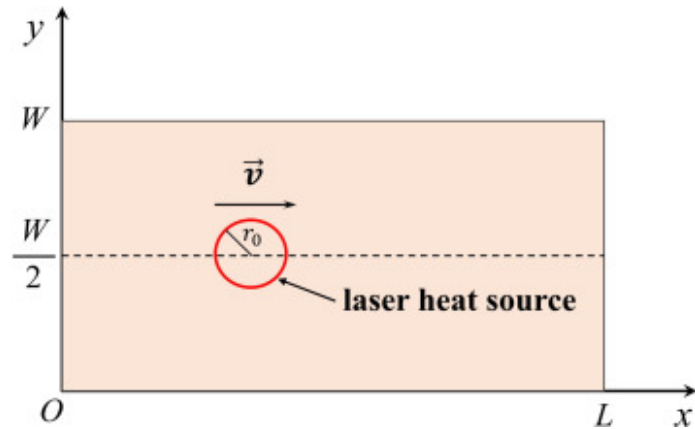
In this section, we consider two-temperature heat transfer in a two-dimensional plate, with the physical geometry illustrated in Fig. 6. Generally, the motion of a moving heat source, such as a laser beam, on the surface of a material can be treated as a transient process. For simplification, this can be modeled as the movement of a point heat source across a rectangular plate. The entire domain has dimensions of $l \times w = 2 \times 1$ and is discretized using a grid of 100×50 particles, allowing for a detailed analysis of the heat transfer dynamics within the plate. This discretization enables us to capture the temperature distribution and or effectively as the heat source moves across the surface.

The numerical simulations are comprised of the following material properties:

$$C_e = 1, C_l = 1, k_e = 1, k_l = 1, \Delta v = 1, 0^T$$

Once H_c and F_T are decided, one can get the relaxation time τ as follows:

$$\tau = \frac{1 - F_T k_e H_c^2}{\|\Delta v\|^2}$$



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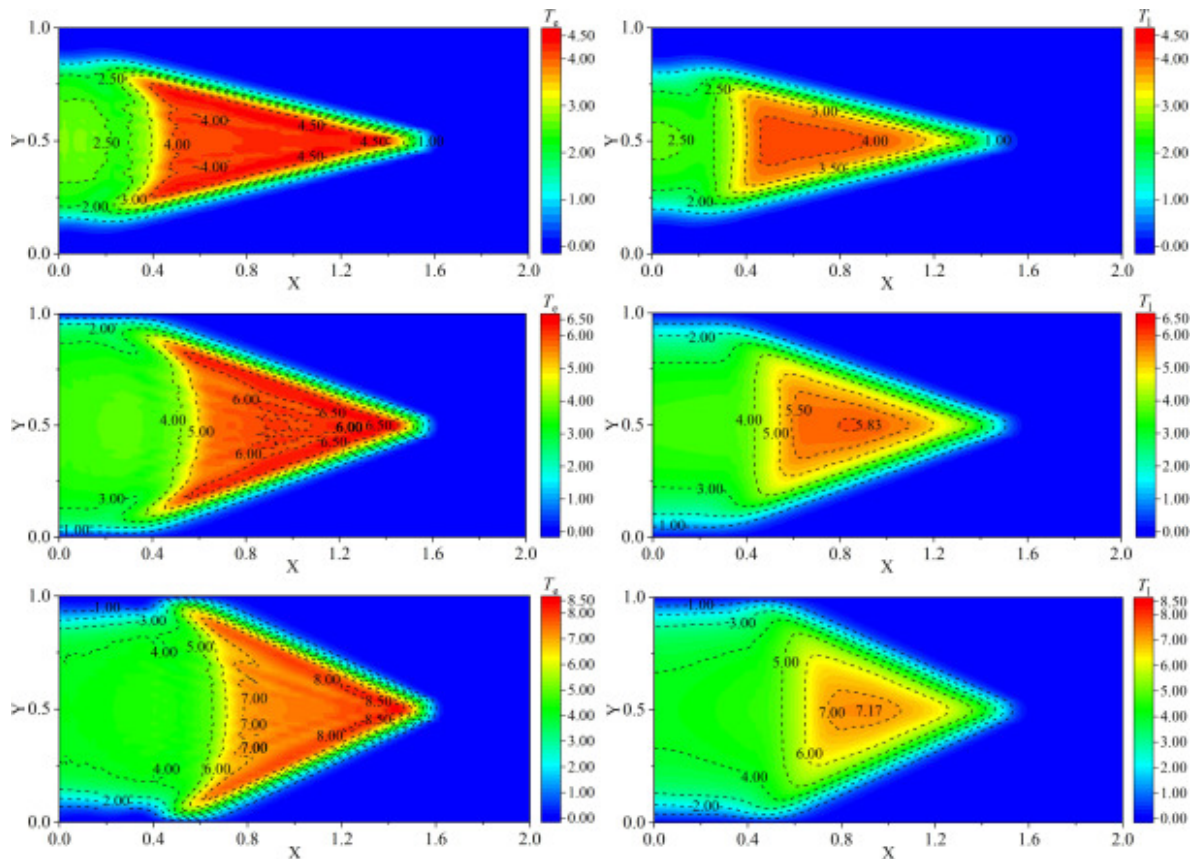
Fig. 6. Two-dimensional model of the plate heated by the moving laser heat source.

4.2.1. Case $F_T = 0$: Pure hyperbolic heat processes

In this case, we analyze the temperature distributions in a pure hyperbolic heat process, characterized by the parameter $F_T = 0$ and a heat source located at $x = 1.6$. The temperature profiles are presented in three different orientations: 30° , 45° , and 60° , with the left column representing the electron temperature T_e and the right column representing the phonon temperature T_l in Fig. 7.

As shown in Fig. 7, the temperature distribution varies significantly at different angles. This variation is due to the differing interactions of electron and phonon within the two-dimensional plate, as well as the position and angular orientation

of the heat source. At a 30° angle, there is a notable increase in temperature near the heat source, forming a distinct temperature peak with a steep gradient. As the angle increases to 45° , the temperature distribution becomes relatively uniform, with a noticeable peak still present, but the variation is less pronounced, and the contour lines are spaced further apart, suggesting a decrease in heat transfer efficiency and a slower rate of heat diffusion. At a 60° angle, the temperature distribution shows broader dispersion, with a significantly lower peak temperature near the heat source, and the contour lines become even sparser, indicating that heat is dissipating over a larger area, leading to increased heat loss efficiency, possibly due to stronger thermal diffusion or convection effects.



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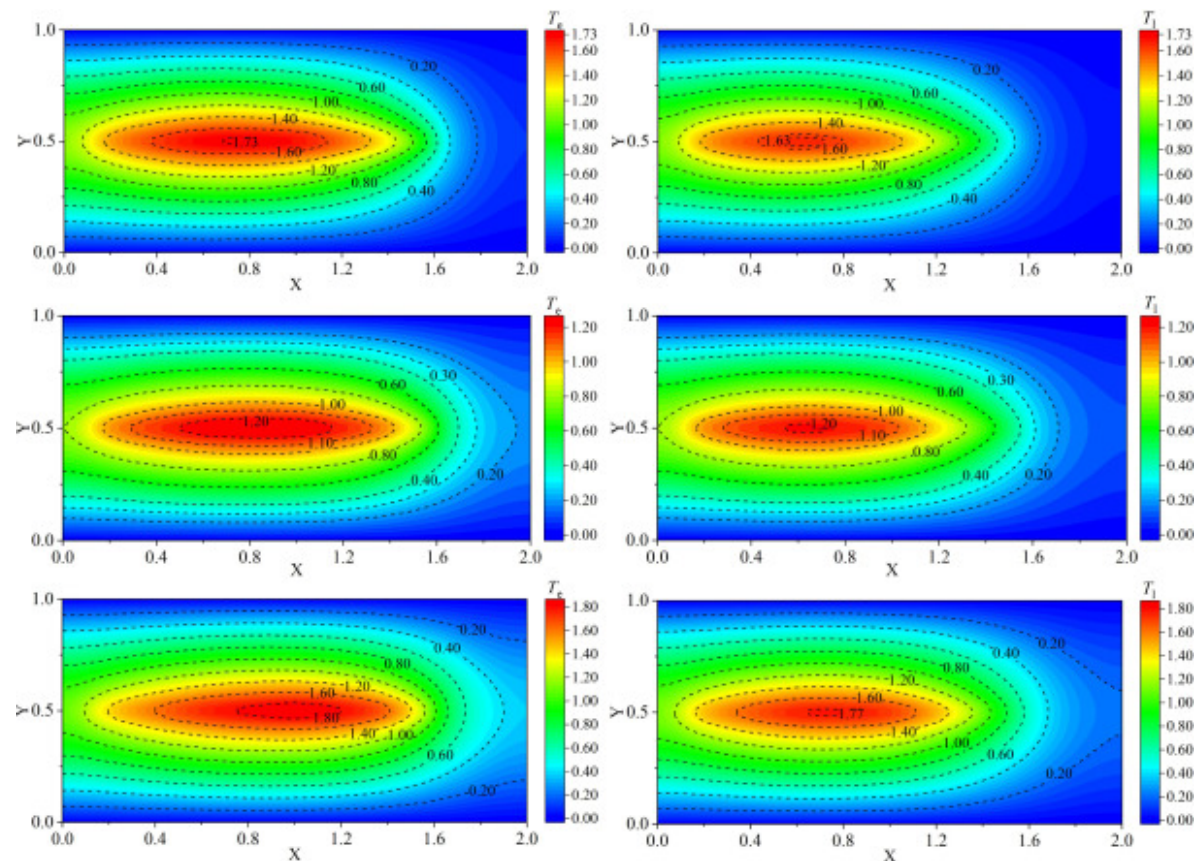
Fig. 7. Temperature distributions with $F_T = 0$ and heat source located at $x = 1.6$. Top: 30° , Middle 45° , and Bottom: 60° . Left column: T_e and Right column: T_l .

A persistent temperature gap ($\Delta T = T_e - T_l$) of 12%–20% is observed across all angles, validating the non-equilibrium nature of energy transfer predicted by the two-temperature model. The divergence is most pronounced near the heat source, where electron excitation dominates. These findings align with the hyperbolic characteristics of the generalized two-temperature model, also emphasizing the crucial role of relative velocity in adjusting the thermal field. Optimizing the velocity ratio through geometric or operational adjustments (e.g., modulating heat source scanning speed) could enable precise control of energy deposition in applications like laser additive manufacturing, minimizing defects caused by thermal stress concentration.

4.2.2. Case $0 < F_T < 1$: General non-Fourier heat processes

In this case, we tune parameter F_T , aiming to investigate the impact of non-Fourier effects on the temperature distribution of electrons and phonons in a two-dimensional plate. These effects significantly alter the temperature distribution, particularly in response to the position of the laser heat source.

Fig. 8 shows the temperature distribution when the laser heat source is located at $x = 0.4, 0.8$, and 1.6 , specifically at the top ($F_T = 0.2$), middle ($F_T = 0.5$), and bottom ($F_T = 0.8$). At the top row, the temperature distribution curve is steep, indicating a faster heat transfer rate with significant non-Fourier effects. The temperature near the heat source is markedly higher than in surrounding areas, showing a strong heat concentration phenomenon. Conversely, at the bottom row, the temperature distribution curve is even flatter, indicating a further reduction in heat transfer rate, approaching the linear heat conduction characteristics described by Fourier's law. The peak temperature near the heat source significantly decreases, with heat dispersing over a larger area. By analyzing the temperature distributions at different F_T values, we conclude that the non-Fourier heat transfer process demonstrates higher efficiency at low F_T values, with heat concentrated near the source. As F_T increases, heat transfer efficiency decreases, and the temperature distribution approaches the linear characteristics of Fourier's law. This highlights the significant role of non-Fourier effects in heat transfer, particularly in scenarios with high temperature gradients and rapid heat transfer.



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Fig. 8. Temperature distributions with different F_T s and heat source located at $x = 1.6$. Top: $F_T = 0.2$, Middle $F_T = 0.5$, and Bottom: $F_T = 0.8$. Left column: T_e and Right column: T_l .

5. Conclusions

In this study, we developed a generalized two-temperature heat transfer model that overcomes the limitations of classical one-temperature models in capturing non-equilibrium heating in metals under rapid thermal processes. By incorporating a moving heat source, we demonstrated the critical role of convection effects arising from the relative motion between the heat source and the metallic body. Our mathematical analyses and simulations revealed hyperbolic characteristics driven by the interplay

between heat transfer speed and heat source velocity, identifying a distinct characteristic angle that governs the thermal response in high-speed heating scenarios, such as advanced manufacturing.

Though this work provides a possible tool for understanding and optimizing heat transfer in metals under a moving heat source, future efforts could focus on developing more robust numerical methods to address the challenges posed by the hyperbolic characteristics of the model [64], [65], [66]. Additionally, while this study considers convection effects induced by the relative velocity between the heat source and the metal, a natural extension would be to explore scenarios involving both a moving heat source and a moving medium, such as pollutant dispersion in flowing water [67]. This advancement could broaden the applicability of the model to a wider range of complex thermal and fluid dynamic systems.

CRedit authorship contribution statement

Wei Wu: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Tao Xue:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Xiaobing Zhang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

None.

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Data availability

No data was used for the research described in the article.

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